

Machine Learning for Forecasting Hazardous Air Quality Days in Indian Cities

Jaykumar Vinodbhai Chauhan

Student ID: 34710280

Supervisor: Dr Tianhai Tian

School of Mathematics, Monash University

MTH5000 Masters Research Project

October 3, 2026

Outline

Introduction

Literature Review

Methodology

Results

Discussion

Conclusion

Outline

Introduction

Literature Review

Methodology

Results

Discussion

Conclusion

Introduction

Placeholder – see report §1.1–1.6.

- ▶ Motivation: hazardous PM2.5 days in Indian cities, and why a forecast is only useful if it arrives before the hazardous day does
- ▶ Research problem, objectives and questions
- ▶ Research significance

Outline

Introduction

Literature Review

Methodology

Results

Discussion

Conclusion

Literature Review

Placeholder – see report §2.1–2.7.

- ▶ Classical and machine learning methods for time series forecasting
- ▶ Deep learning: RNNs, vanishing gradients, the GRU
- ▶ Prior work on Delhi / Indian air quality
- ▶ Forecast verification as a warning system; data quality in reference monitors
- ▶ The research gap this project fills

Outline

Introduction

Literature Review

Methodology

Results

Discussion

Conclusion

Methodology – Data

Placeholder – see report §3.1–3.3.

- ▶ Data collection and station selection
- ▶ Preprocessing: two defects fixed in the source data
- ▶ Seasonality analysis and feature construction
- ▶ Weather covariates

Methodology – Models

Placeholder – see report §3.4–3.5.

- ▶ Classical benchmark: ARIMA with Fourier terms
- ▶ Penalised linear regression, random forest, histogram gradient boosting
- ▶ The GRU: architecture and training

Methodology – Evaluation

Placeholder – see report §3.6–3.9.

- ▶ Rolling-origin evaluation harness
- ▶ Evaluation measures
- ▶ Statistical comparison of forecasts (Diebold–Mariano)
- ▶ Reproducibility

Outline

Introduction

Literature Review

Methodology

Results

Discussion

Conclusion

Results

Placeholder – see report §4.1–4.3.

- ▶ Forecast accuracy across horizons (Table 4.1)
- ▶ Does weather change the answer?
- ▶ Two arms of the same question: regression vs. classification
- ▶ The decision threshold

Outline

Introduction

Literature Review

Methodology

Results

Discussion

Conclusion

Discussion

Placeholder – see report §5.1–5.3.

- ▶ The operational question dominates the modelling question
- ▶ How large is a difference between models?
- ▶ Limitations

Outline

Introduction

Literature Review

Methodology

Results

Discussion

Conclusion

Conclusion

Placeholder – see report §6.1–6.3.

- ▶ Summary of the study
- ▶ Main conclusions
- ▶ Future work

Questions?